ABSLD-DF: Demographic- Group Fairness in Adversarially Robust Knowledge Distillation

CS621 Course Project

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The Core Dilemma: ARD vs Unfairness

Adversarial Robustness

Adversarial Robustness Distillation (ARD) produces highly defensive networks that protect AI against malicious input perturbations.
Goal: Ensuring model reliability in high-stakes environments.

Robust Unfairness

Security is often unequally distributed. Models become intensely secure on "**easy**" classes but remain highly vulnerable on "**hard**" classes.
Challenge: How do we build defenses that protect all classifications equally?

Ethical Blindspot in Base Models

The ABSLD Baseline

The Anti-Bias Soft Label Distillation (ABSLD) framework successfully addresses semantic unfairness (e.g., Cat vs Bird).

The Gap: It lacks mechanisms to detect demographic unfairness. If a Teacher model harbors biases (gender or skin tone), the Student blindly inherits them.

- Semantic focus ignores human elements
- Biases are amplified during distillation
- Critical risk for facial verification AI

Comparison: ABSLD vs ABSLD-DF

Dimension	Original ABSLD	Our ABSLD-DF
Unit of Fairness	Semantic class (cat, dog...)	Demographic group (Gender, Skin Tone...)
Error Calculation	Class-wise CE loss	Group-wise CE loss (Val & Test)
Dataset focus	CIFAR-10 (No humans)	CelebA / FairFace (Faces + Labels)
Ultimate Goal	Equal object robustness	Equal human-group robustness

Decoupling the primary task from sensitive attributes ensures model security never induces discrimination.

The 4-Stage ABSLD-DF Framework

Stage 1

Datasets & Pipeline Creation focusing on CelebA and FairFace.

Stage 2

Pre-Distillation Teacher Audit for baseline transparency.

Stage 3

Demographic Temperature Control for fairness scaling.

Stage 4

Multi-Stage TA pipeline for smoother transfer.

Stage 1.0: Datasets

Celeba

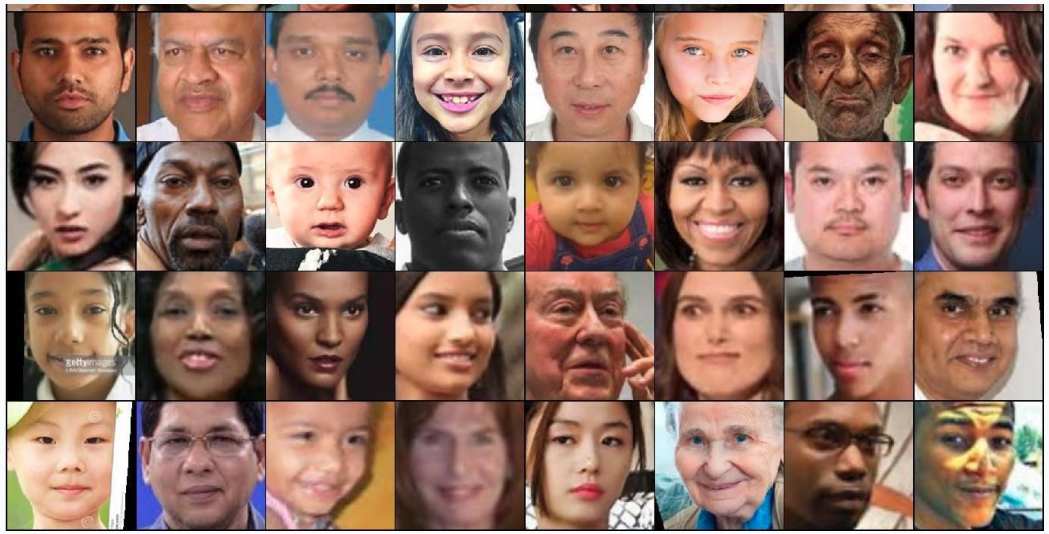
- **~202,599 images**
- **~10,177 identities**
- **40 binary attributes** per image
  (e.g., *Male, Young, Smiling, Glasses*)

FairFace

- **~108,501 images**

Labels:

- **Race (7 classes)**
- **Gender (2 classes)**
- **Age (9 groups)**



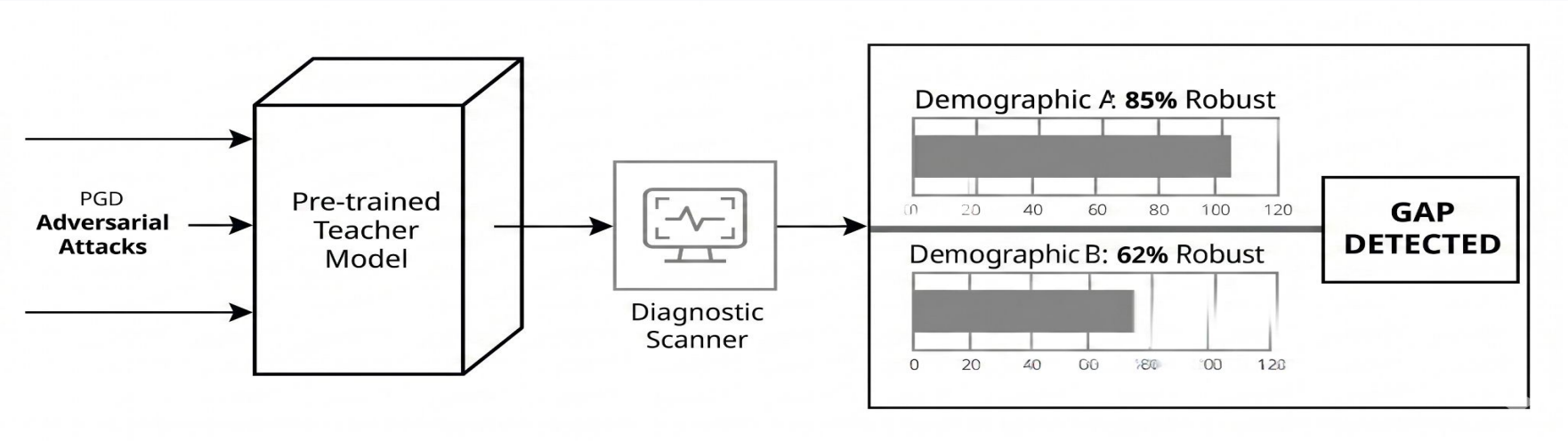
Stage 1.1: Teacher Training

Model Selection

We run **Projected Gradient Descent (PGD)** attacks on the various models to identify a suitable teacher model.

Model	Clean (F1)	Robust (F1)
WideResNet-34-10	0.53	0.52
ResNet-50	0.90 0.82	0.52 0.58
ResNet-34	0.90 0.89	0.57 0.63
ResNet-18	0.90 0.89	0.57 0.62
MobileNet-Large	0.90 0.89	0.46 0.50
MobileNet-Small	0.90 0.88	0.34 0.39

Stage 2: Pre-Distillation Audit



Transparency via Audit

We run **Projected Gradient Descent (PGD)** attacks on the pre-trained Teacher model to measure its demographic robustness gaps.

Stage 2: Pre-Distillation Audit

ResNet-18: Clean F1: 89.09, Robust F1: 60.41

Sensitive	Pos N	Neg N	Pos Clean F1	Neg Clean F1	Pos Robust F1	Neg Robust F1	F1 Gap
Mouth_Slightly_Open	9573	10294	95.09	66.3	**78.54**	**10.34**	68.2
High_Cheekbones	8926	10941	93.15	71.69	**75.21**	**20.58**	54.62
Eyeglasses	1383	18484	84.1	89.39	**39.6**	**61.71**	22.12
Male	8458	11409	86.77	90.41	**49.96**	**66.71**	16.75
Gray_Hair	967	18900	86.03	89.26	**45.9**	**61.25**	15.35
Blond_Hair	3056	16811	90.73	88.7	**68.51**	**58.55**	9.95
Chubby	1216	18651	87.18	89.24	**52.36**	**61.04**	8.68
Bald	411	19456	89.09	89.09	**52.27**	**60.61**	8.33
Young	14832	5035	89.08	89.09	**62.45**	**55.35**	7.1
Pale_Skin	856	19011	88.37	89.11	**55**	**60.58**	5.58
Big_Nose	4943	14924	91	88.22	**64.25**	**58.72**	5.52
Brown_Hair	4793	15074	89.85	88.84	**61.85**	**59.94**	1.91

Stage 3: Temperature Control

Mechanism

Dynamically lower the temperature for vulnerable groups. This produces "sharper" supervision soft labels , forcing the model to intensely focus on correcting its demographic blindspots.

z = [3.0, 0.5, 0.2, 0.1] (class 1 is correct)

τ = 1.0 (normal): → soft label ≈ [0.87, 0.07, 0.04, 0.03] — fairly sharp

τ = 0.5 (lower): → soft label ≈ [0.98, 0.01, 0.01, 0.00] — very sharp

τ = 5.0 (higher): → soft label ≈ [0.40, 0.23, 0.20, 0.17] — very smooth

$$\tilde{\tau}_g \leftarrow \tilde{\tau}_g - \beta \cdot \frac{R_g - \bar{R}}{\max_g |R_g - \bar{R}|}$$

Sharper Supervision

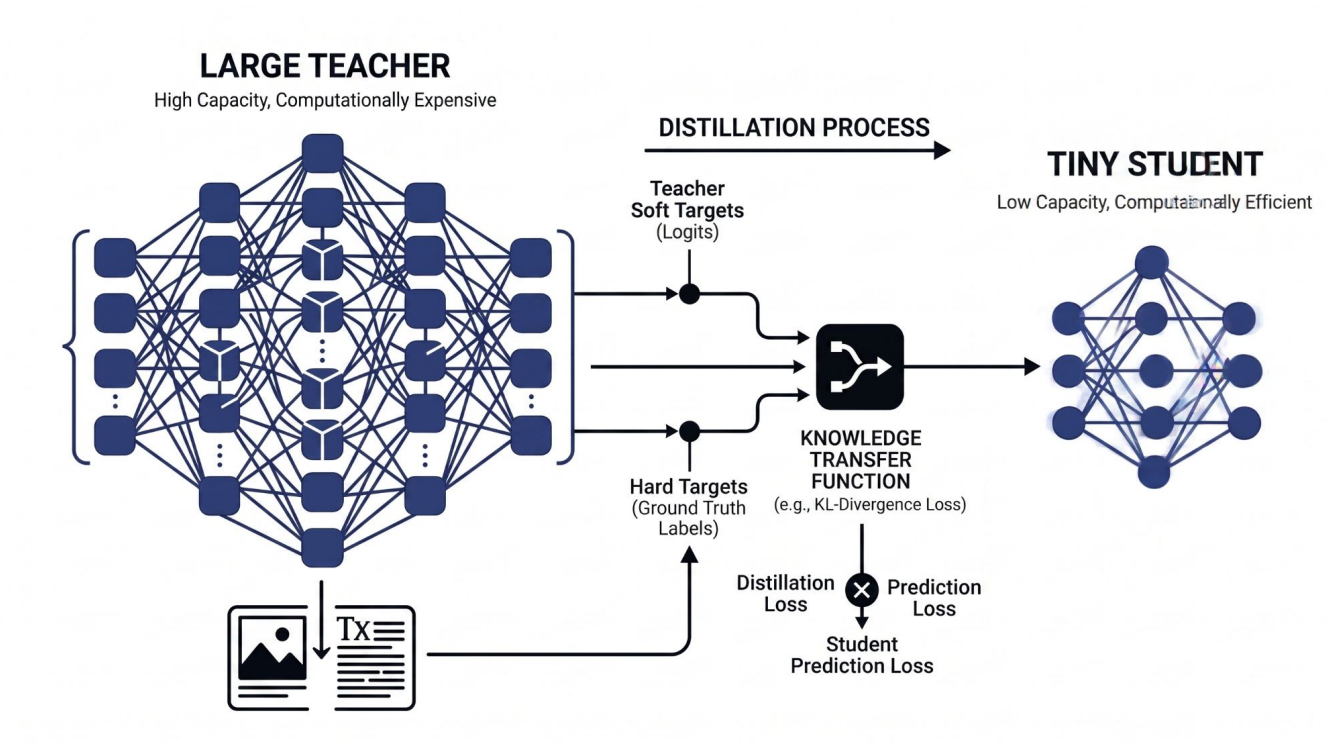
Forces gradients to prioritize group 'g' error correction.

The Capacity Gap Problem

The Hazard

A massive capacity gap between a Large Teacher and a Tiny Student causes the Teacher to give misleadingly confident soft labels early in training.

Theorem 2 requires the Teacher to have a more correct prediction distribution than the Student across ALL groups to guarantee fairness.



Stage 4: The Teacher Assistant Bridge

Teacher

Large Network (Source of robust knowledge)

Standard KD

Initial capability transfer phase

TA Bridge

Intermediate model reduces capacity gap

Student

Final robust and fair tiny model

Because the TA is closer in capacity to the Student, its demographic error signals remain trustworthy throughout the entire training loop.

Stage 4: The Teacher Assistant Bridge

Model			Clean F1	Robust F1
Teacher	**ResNet-18**	Full Training	0.89	0.62
TA	**MobileNet-Large**	Full Training	0.89	0.50
		KD	0.88	0.54
Student	**MobileNet-Small**	Full Training	0.89	0.38
		KD	0.86	0.43
		KD + Temperature	0.87	0.40

	Orignial			**Our**		
Target	**Pos**	**Neg**	**Diff**	**Pos**	**Neg**	**Diff**
Mouth Open	59.31	2.4	56.9	55.6	10.4	44.2
Male	26.67	46.17	19.5	29.33	43.48	14.14
Young	40.21	34.75	5.45	38.18	32.77	5.41

Project Outcomes

ABSLD-DF Framework

Training pipeline using demographic scaling on CelebA.

Fairness Audit Tool

Standalone script to evaluate biases in pre-trained Teachers.

Report

Comprehensive report detailing results and limitations.

Questions?

Thank you for your attention.

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